ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B3 · CHANGING SURFACE

Shaping the land

★☆☆
Grade 5

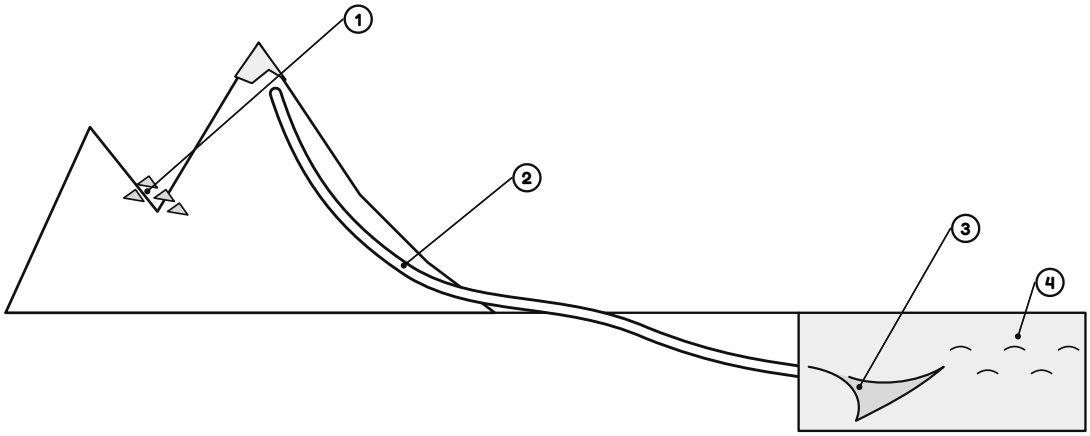
Learning goals

- Label the processes.
- Sort fast and slow changes.

1 Label the diagram. [4]

LABELS

deposition (delta forms) **weathering (rock breaks up)** **sea**
erosion (river carries rock and soil)



1 weathering (rock breaks up)

2 erosion (river carries rock and soil)

3 deposition (delta forms)

4 sea

2 Match each word to its meaning. [5]

Draw a line to match each pair, or write the letter.

weathering ●	→ E	● **A** sand and silt being dropped
erosion ●	→ D	● **B** land built where a river meets the sea
deposition ●	→ A	● **C** sea water pushed onto the land by a cyclone
delta ●	→ B	● **D** rock and soil being carried away
storm surge ●	→ C	● **E** rocks breaking into pieces

3 **Sort the changes into slow and sudden.** [6]

SORT THESE

rock weathering in a cliff **a delta growing** **an earthquake** **a cyclone storm surge** **a river valley getting deeper** **a landslide**

Slow changes	Sudden changes
a delta growing	an earthquake
rock weathering in a cliff	a landslide
a river valley getting deeper	a cyclone storm surge

4 **Fill in the gaps.** [3]

WORD BANK

monsoon **winter** **desert** **tectonic** **delta**

Bangladesh is built on the world's largest ___delta___. Rivers flood most during the ___monsoon___. The Earth's outer layer is made of ___tectonic___ plates.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B3 · CHANGING SURFACE

Causes and patterns

★★☆
Grade 5

Learning goals

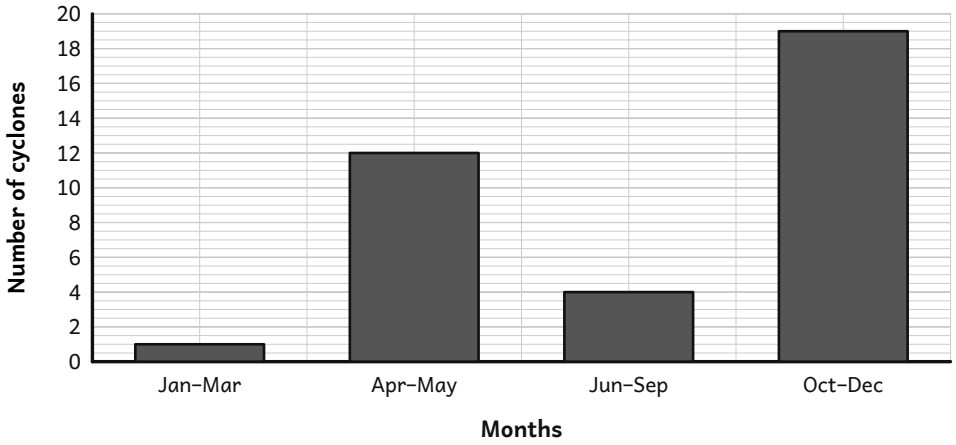
- Explain causes of change.
- Use a pattern to make predictions.

1 **Explain how a river can both destroy land and make new land.** [2]

Fast water erodes banks and carries soil away; where the river slows, it deposits the soil and builds chars and the delta.

2 **This table shows (example) the number of cyclones that formed in the Bay of Bengal in each season over 20 years.** [5]

Months	Jan–Mar	Apr–May	Jun–Sep	Oct–Dec
Cyclones	1	12	4	19



a) Draw the bar chart.

b) Describe the pattern.

Most cyclones form in April–May and October–December; very few form in January–March.

c) In which months should coastal people be most prepared?

April–May and October–November/December

3 **Why are landslides more likely on hills where the trees have been cut down?** [2]

Tree roots hold soil in place and leaves slow the rain; without them, heavy rain soaks and loosens the soil so it slides.

4 **Floods are dangerous, but farmers in Bangladesh also say "the flood brings gold". What do they mean?** [2]

Floods deposit fertile silt on the fields, which helps crops grow well afterwards.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

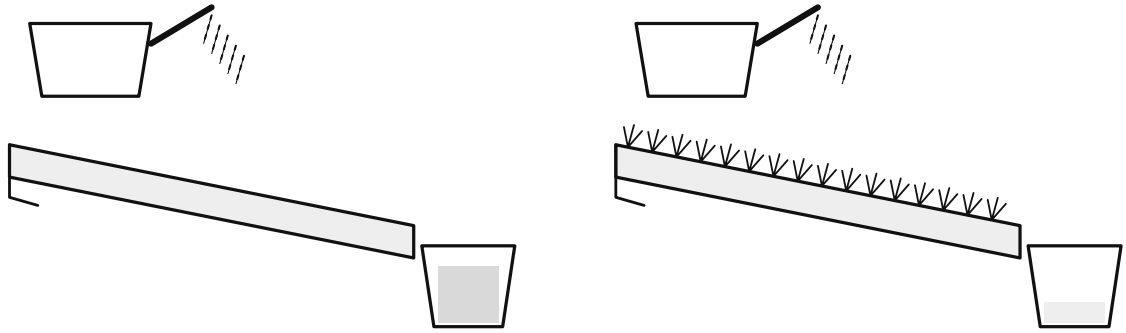
INVESTIGATE · B3 · CHANGING SURFACE

Do plants stop soil erosion?

★★☆
Grade 5

Learning goals

- Use a model to investigate erosion.
- Control variables.
- Draw conclusions and evaluate the model.



bare soil **soil with grass**

Two trays at the same slope. The same amount of water is poured on each.

Class 5D tested two trays: one of bare soil and one where grass had grown for three weeks. They poured 500 ml of water from the same height onto each tray and weighed the soil washed into the cup. They repeated it three times.

1 **Complete the table and find the averages.** [2]

Tray	Test 1 (g)	Test 2 (g)	Test 3 (g)	Average (g)
bare soil	62	58	60	60
soil with grass	9	12	9	10

2 **List three things they kept the same.** [3]

slope (angle) of tray, amount of water, height it was poured from, type and amount of soil, size of tray

3 **Write a conclusion.** [2]

Grass greatly reduces erosion: about 6 times less soil washed away (10 g compared with 60 g).

4 **Explain your conclusion using what roots and leaves do.** [2]

Roots hold the soil together; leaves and stems slow down the water so it carries less soil.

5 **How is this model like a real hillside in the Chattogram hills, and how is it different?** [2]

Like: soil on a slope, rain washing it down, plants holding it. Different: much smaller, the real hill has rock, trees and much more rain over a long time.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B3 · CHANGING SURFACE

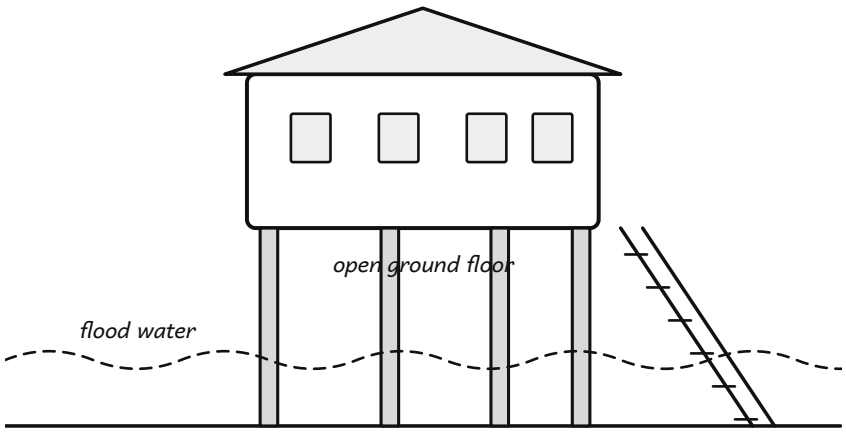
Ready for the storm: saving lives from cyclones

★★★
Grade 5

Learning goals

- Use data to show change over time.
- Explain how science and people reduce risk.
- Suggest further actions.

Science | Geography | History | Maths | Citizenship



A cyclone shelter on raised pillars, used as a school in normal times.

In November **1970** the Bhola cyclone struck the coast of what is now Bangladesh. With almost no warning and few safe buildings, an estimated **300 000 to 500 000** people died, mostly drowned by the storm surge.

After this disaster, the **Cyclone Preparedness Programme** began in 1972. Today tens of thousands of trained volunteers warn villages using radios, megaphones and flags, and help people reach thousands of raised **cyclone shelters**. Weather scientists use satellites to track storms days before they arrive.

When Cyclone **Sidr** hit in 2007, about **3400** people died: still a terrible loss, but far fewer than in 1970. When Cyclone **Amphan** hit in 2020, more than 2 million people were moved to shelters and fewer than 30 people died in Bangladesh. The mangroves of the Sundarbans also helped to slow the wind and waves.

Sources: Bangladesh Cyclone Preparedness Programme; UN and Red Cross reports. Death tolls are estimates.

1 **Complete the table.** [3]

Cyclone	Year	Approximate deaths in Bangladesh
Bhola	1970	300 000 – 500 000
Sidr	2007	about 3400
Amphan	2020	fewer than 30

2 **Describe the pattern in the deaths over time and give three reasons for it.** [4]

Deaths fell hugely. Reasons: early warnings from satellites and forecasts; volunteers spreading warnings; cyclone shelters; evacuation of millions; mangrove protection; better roads and awareness.

3 **Why are cyclone shelters built on tall pillars with an open ground floor?** [2]

So the storm surge and flood water can flow underneath while people stay safe above the water.

4 **Some families do not want to leave their homes, animals and belongings. Suggest how volunteers could persuade them, and how shelters could be improved.** [2]

Explain the danger clearly; leave early; allow animals near shelters; separate safe spaces for women and girls; water, toilets and food in shelters; protect property.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · B3 · CHANGING SURFACE

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

G	O	J	V	T	R	W	J	W	A	B
F	N	N	C	C	Y	C	L	O	N	E
L	K	R	H	E	R	O	S	I	O	N
O	K	U	I	Y	E	O	W	T	S	L
O	T	P	B	V	P	J	M	D	T	S
D	V	D	J	J	E	L	S	R	J	V
R	A	E	I	M	K	R	A	N	W	I
O	L	L	T	D	T	O	S	T	U	P
C	C	T	I	P	F	P	W	I	E	T
K	N	A	M	F	O	W	P	O	L	E
U	E	B	C	H	A	R	T	C	S	T

char **cyclone** **delta** **erosion** **flood** **plate** **river** **rock** **silt**

2 **Which process drops sand and silt to form a delta?** [1]

- ○ weathering
- ○ erosion
- ● **deposition**
- ○ evaporation

3 **What causes earthquakes?** [1]

- ○ heavy rain
- ● **tectonic plates suddenly moving**
- ○ strong wind
- ○ the Moon

4 **Sort these into helpful and harmful effects of a flood.** [5]

SORT THESE

new silt for fields **fills ponds with fish** **homes washed away** **crops destroyed** **dirty water spreads disease**

Helpful	Harmful
new silt for fields	crops destroyed
fills ponds with fish	homes washed away
	dirty water spreads disease

5 **Challenge. Why do you think Bangladesh has more new land (chars) in some places and loses land in others, even in the same year?** [2]

Rivers erode banks where the water is fast (outside of bends) and deposit silt where it slows (inside of bends, near the sea).